# Yuhan Chen

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## PROFESSIONAL SUMMARY

- Proficient in a wide range of programming languages and software tools, including C, Python, PyTorch, MATLAB, Java, Scikit-learn, TensorFlow, Android Studio, and Simulink, with experience developing on Linux systems using VMware virtual machines

## EDUCATION

**Southern University of Science and Technology (SUSTech)** ***Shenzhen, China***

**Major:** Communication Engineering **Degree Expected:** Bachelor of Engineering **GPA:** 3.7/4.0; 89.42/100 *08/2022-07/2026*

**Coursework:** Calculus, College Physics, Introduction to C Programming, Engineering Mathematics, Analog Circuits, Analog Circuits Laboratory, Digital Circuits Laboratory, Signals and Systems, Engineering Electromagnetics, Advanced Electronic Science Experiment, Probability and Statistics, Data Communications and Networking, Microprocessors and Microsystems, Wireless Communications, Microwave Engineering, Digital Signal Processing, Optoelectronics IntelliSense, Antennas and Radio Propagation, Solid-state Electronics, Digital System Design

## CONFERENCE PUBLICATION & PAPER

*Performance Analysis of Distance-Based Frequency Allocation in a Multi-Cell Cellular Network* *10/2025*

Advisor: Dr. Kai-Kit Wong, Electronic and Electrical Engineering Department, University College London (UCL), The 2nd International Conference on Computer Communication, Networks and Information Science (CCNIS 2025), as the first author

## WORK EXPERIENCE

**Hardware R&D Intern, HDL Automation CO., LTD, Guangzhou, China** *07/2025-08/2025*

- Participated in end-to-end hardware development and testing workflows, proficiently operated instruments including multimeters, oscilloscopes, vector network analyzers, and spectrum analyzers to conduct functional testing, performance validation, and fault diagnosis, and produced well-structured test reports to support hardware iteration
- Developed strong electronic assembly and debugging capabilities, proficient in soldering and rework of 0402 SMD components; contributed to the identification and mitigation of spurious emissions in a ZigBee antenna by optimizing impedance matching to improve RF performance
- Collaborated effectively with cross-functional teams including hardware, software, and testing engineers, clearly reported progress in project meetings, analyzed technical issues during discussions, and implemented improvement actions in a timely manner

## RESEARCH EXPERIENCE

*Performance Analysis of Distance-Based Frequency Allocation in a Multi-Cell Cellular Network* *04/2025-10/2025*

- Modeled and simulated multi-cell cellular networks using distance-based frequency allocation, representing base stations as uniformly spaced hexagonal grids in Cartesian coordinates and incorporating path-loss and Rician fading to compute signal-to-interference ratios
- Conducted large-scale simulations across network configurations ranging from 7 to 61 cells with reuse factors from 7 to 36, systematically analyzing the impact of reuse factor and network size on average channel capacity
- Identified non-monotonic capacity trends, showing that with larger networks the average channel capacity initially decreases and then increases as the reuse factor grows, and proposed an expanded cellular network framework to support higher reuse factors and future integration of advanced allocation algorithms

## ACADEMIC PROJECTS

**Wireless Communication System Designing, Southern University of Science and Technology (SUSTech)** *12/2024-01/2025*

- Used HFSS and ADS to design, simulate, and optimize antennas, power dividers, and amplifiers, and fabricated the designs on printed circuit boards
- Worked with LabVIEW and USRP to develop modulation schemes and signal processing modules for the system

**mmWave Radar for Distance Detection, Southern University of Science and Technology (SUSTech)** *09/2024-12/2024*

- Utilized a Vector Network Analyzer and antennas to build a mmWave radar by detecting phase changes of scattering parameters (S21), which indicated the movement of target objects.
- Applied the radar to detect throat movements and identify different tones in Mandarin using an MLP algorithm

**Embedded Control System Designing, Southern University of Science and Technology (SUSTech)** *07/2024-08/2024*

- Designed an autonomous line-following car for the National Undergraduate Electronics Design Contest, using sensors and gyroscopes with PID control algorithms to ensure precise tracing
- Awarded the second prize of Guangdong Province

## EXTRA-CURRICULAR ACTIVITIES

**Community Worker, Nantou Sub-district Service Center, Shenzhen, China** *05/2024-08/2024*

- Participated in community activities, including regular facility inspections, mediating neighbor disputes, and supporting elderly residents
- Organized and conducted a science workshop for local elementary students, from material preparation to lesson delivery, focusing on hands-on experiments to explain atmospheric pressure
- Awarded the "Excellent Team Second Prize" and "Outstanding Individual" for contributions to community work

**Deputy Leader, Student Club, Zhiren Chinese Orchestra, Southern University of Science and Technology (SUSTech)** *09/2023-09/2024*

- Coordinated on-site operations, ensuring smooth execution of live events through effective timing, communication, and real-time problem-solving during rehearsals and performances
- Organized multiple university-level music performances with over 100 participants and 300+ attendees, overseeing all logistical aspects including stage management, rehearsal coordination, and personnel scheduling

## HONORS & AWARDS

Second Prize in Guangdong Province, National Undergraduate Electronics Design Contest *08/2024*

Third-class Scholarship, University-level, Southern University of Science and Technology (SUSTech) *09/2023*

## SKILLS & INTERESTS

- Language: English (fluent), Chinese - Mandarin (native)
- Interests: Music (Piano, Guitar and Drums), Travelling, Gym Training, Motorsports